

Supplementary Information: Recursive operand closure and certified spanning generate a cognitive near-singularity

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Contents

1 Purpose, status and closure boundary	2
2 Formal objects and type discipline	2
2.1 Functional state spaces and carriers	2
2.2 Conceptual basis and strict self-typing	2
2.3 Three-dimensional participant fitness	3
3 Canonical registry, aliases and obligations	3
4 Coupled state, generator and event semantics	4
4.1 Conformance and replay	4
5 Deterministic witnesses and metamorphic tests	4
6 Synthetic world generator and intervention patches	5
6.1 Regime coordinates	5
6.2 Paired nonfactorizing target	5
6.3 Ten conditions	5
7 Integrated CNS results	6
8 Option containment, governance and hazard projection	6
9 Bounded registration and correlation-aware recovery	6
10 Model codewords and held-out identification	7
11 Artifact reconstruction and propagation status	8
12 Statistical plan and uncertainty	8
13 Counterexample and non-dominance regimes	8
14 Claim-status ledger	9
15 Complete generated registry and result tables	9
16 Reproducibility manifest	11
17 External completion protocol	14

1 Purpose, status and closure boundary

This Supplement defines the complete finite computational object, states the theorem targets used by the Article, records the canonical function registry, gives the deterministic witnesses and experiment configuration, and reports all executed synthetic results. It is designed to be read without an unpublished dependency.

The execution reached two of the three preregistered closure targets. Formal closure is supported within the declared finite witness, target, budget and resolution classes. Execution closure is supported by one generator, one model hash, deterministic replay and read-only result queries. Propagation closure is UNDETERMINED. The available environment did not provide three independent external language-model families in fresh contexts, externally available reasoning traces from three version-locked model families, or an independent historical audit. The algorithmic artifact-reconstruction experiment in Section 11 is a structural stress test and is not substituted for those missing boundaries.

The status-conditioned title is therefore Status 3: *Recursive operand closure and certified spanning generate a cognitive near-singularity*. The Article makes no claim that the model is the historical Great Filter, that a CNS is civilization’s highest-importance innovation, or that the bounded-observation parameters describe any current AI system.

2 Formal objects and type discipline

2.1 Functional state spaces and carriers

Definition 1 (Functional state-space instance). *An FSS instance X is a represented state domain equipped with a declared internal operator basis, targets, admissibility rules and a transition semantics. The modeled collection is*

$$\mathfrak{X} = \{\mathcal{C}\} \cup \{\mathcal{F}_i : i \in I\}, \quad (1)$$

where $\mathcal{C}$ is conceptual and $\mathcal{F}_i$ is participant i ’s fitness FSS.

Definition 2 (Typed signal carrier). *For each $X \in \mathfrak{X}$, Σ_X is a typed carrier with source, destination, encoding, capacity, delay, channel state and provenance. It has no internal FSS basis and is invalid if assigned a recursively generated signal boundary.*

This distinction ends the signal-space regress. A carrier can be transformed, erased, delayed or compressed without becoming a second reasoning domain. Shannon-style channel structure motivates the carrier object, but the target-relative contracts below are internal definitions[1, 2].

Definition 3 (Boundary duality). *Register $_X : \Sigma_X \times X \rightarrow X$ admits or rejects typed signal structure into X . Express $_X : X \rightarrow \Sigma_X$ encodes selected represented structure onto a carrier. Exact round-trip identity is not required for every target; the contract may preserve identity, relation, verdict or action at a declared resolution.*

Proposition 1 (Bidirectional necessity for observable interaction). *For a target class containing both externally grounded distinctions and externally checkable consequences, any observable interacting FSS requires target-distinguishable input and output transduction.*

Proof. Remove input transduction. Then there exist external carrier states with distinct target consequences that induce the same FSS state, so the input target cannot factor through the interaction. Remove output transduction. Then there exist FSS states with distinct externally checkable consequences that produce the same carrier state, so the output target cannot factor through the interaction. The conclusion is target-class conditional: an isolated system or a one-direction-only target lies outside the premise. $\square$

2.2 Conceptual basis and strict self-typing

The conceptual domain is a finite typed multigraph with node levels, relation types, boundary conditions, active context, recoverable identities, evidence, verdicts and obligations. Level one contains grounded nodes. For $n \geq 1$,

$$X_{n+1} = \{\rho_n(A) : A \in \text{AdmSub}_n(X_{\leq n})\}, \quad (2)$$

where AdmSub_n is the frozen admissibility predicate and ρ_n is Create/promotion. Type and boundary maps are strictly level-raising; impredicative self-application is rejected.

Definition 4 (Store and Recall). *Store preserves a target-recoverable identity, relation, verdict or obligation with provenance. Recall reactivates a stored object under an active target and context budget. Persistence and reactivation are distinct contracts: a stored object may not be recalled, and an active object may have no recoverable stored provenance.*

Definition 5 (Local and global assessment). *L evaluates or transforms within the active represented region. G may traverse represented regions and must preserve every successor obligation required by the declared global target. Generic connectivity is not sufficient for G.*

Proposition 2 (Conceptual basis within the declared target family). *Given grounded nodes supplied by Register and admissible composites supplied by Create/promotion, the internal basis {Store, Recall, L, G} is irreducible for the target family in Table 4.*

Proof. Witnesses W03–W06 provide a target pair for each contract. Removing Store loses persistent identity after context eviction. Removing Recall leaves a stored bridge inactive. Removing L prevents the local evidence verdict. Removing G makes two globally different support graphs locally indistinguishable. The hidden-resource suite prohibits an external index, evaluator label or precomputed global target from supplying the removed contract. Irreducibility is not asserted beyond this exhibited family. $\square$

2.3 Three-dimensional participant fitness

For participant i , target τ and candidate action a ,

$$\mathbf{f}_{i,t}(a) = \left(f_{i,t}^{\text{cur}}, f_{i,t}^{\text{tar}}, f_{i,t+1|a}^{\text{proj}} \right). \quad (3)$$

The coordinates may be Euclidean only when the declared target order has a valid embedding $u_{i,\tau}$. A non-embeddable partial order remains a partial order and produces UNDETERMINED rather than a fabricated scalar.

Theorem 1 (Three-coordinate necessity and sufficiency for the declared controller). *Current, target and action-conditioned projected fitness are independently necessary and jointly sufficient for the finite action-selection target family used in the benchmark.*

Proof. For each omitted coordinate, the benchmark contains a paired family in which the retained two coordinates are identical and the optimal action is opposite. Any deterministic two-coordinate controller is therefore at most 0.5 accurate on that designated pair. Given all three coordinates, the controller evaluates current viability, target adequacy and action-conditioned successor adequacy and resolves every pair. Sufficiency is relative to the declared target family, exact projection and admissibility gate. $\square$

The three pairwise relations induce six functions:

Relation	Forward contract	Complementary contract
Current–projected	Model	Evaluate
Current–target	Stability	Adaptation
Target–projected	Decomposition	Bridging

Model represents action-conditioned consequences; Evaluate compares them with target and viability. Stability preserves a viable path; Adaptation changes a failing one. Decomposition factors a failed target into typed obligations; Bridging constructs an admissible correction across them. The deterministic suite contains a matched target pair for every function.

3 Canonical registry, aliases and obligations

The registry is the source of truth for function identifiers, display names, layers and contracts. Every local inventory is generated from this object. Admissible aliases may change notation but not domain, codomain, target or transition contract.

Two residual obligations are retained in `registry/obligations.json`. R1 records that Store was omitted from an earlier presentation and is closed by the canonical registry plus projection checks. R2 records that Destroy is not included in the minimal basis because no independent target-pair witness survives the registered composite operations; its status remains bounded-open.

4 Coupled state, generator and event semantics

The complete state is

$$Z_t = \left(\{X_t, \Sigma_{X,t}, \text{Register}_{X,t}, \text{Express}_{X,t}\}_{X \in \mathfrak{X}}, \Gamma_t, o_t, M_t, \{\mathbf{f}_{i,t}, \hat{\mathbf{f}}_{i,t}\}_{i \in I}, \Lambda_t, e_t, \mathcal{R}_t \right), \quad (4)$$

where Γ_t is active context, o_t the focal operand, M_t the recoverable registry, Λ_t the typed coupling ledger, e_t environment state and $\mathcal{R}_t$ resource counters. The only primary transition is

$$\text{step}(Z_t, a_t, \xi_t) \mapsto (Z_{t+1}, E_{t:t+1}). \quad (5)$$

Every trajectory executes the following dependency order:

1. Register grounded signal structure.
2. Recall or activate represented operands.
3. Apply L or G , optionally construct a lifted operand and a global span.
4. Validate a target-relative certificate.
5. Model and Evaluate participant consequences, Decompose residuals, Bridge admissible corrections, and apply Governance.
6. Store accepted identities, verdicts and obligations and test recursive reuse.
7. Express the target-relevant state through the output carrier.
8. Realize the admitted consequence and close the event batch.

Each event stores trajectory and event identifiers, operator, model and registry hashes, target, seed, resource and residual deltas, and pre/post state hashes. The final execution contains 12800 trajectories and 102400 events. Every row carries model hash prefix `797b16e30934` and registry hash prefix `b7a6ef0e1d8c`.

4.1 Conformance and replay

The executable conformance suite passed all checks:

- registry identifiers were unique and R1/R2 were present;
- signal carriers validated as non-FSS objects;
- trajectory and event tables contained one model and registry hash;
- each trajectory contained exactly eight events;
- regeneration of the first trajectory from its regime and seed reproduced every post-state hash and all selected summary fields;
- no designated ablation passed its target;
- signal-regress and split-generator codewords differed from the canonical codeword.

The exact machine-readable report is `tests/conformance_results.json`.

5 Deterministic witnesses and metamorphic tests

Table 4 lists all 23 target pairs. The full model passed each row. Ablations returned FAIL where the target could be falsified directly and UNDETERMINED where the removed distinction made a verdict unavailable. No statistical test was used to rescue a failed deterministic contract.

Metamorphic transformations preserving target verdicts were admissible renaming, node relabeling, graph isomorphism, order-preserving transformations of embedded fitness, event replay and contract-preserving inventory grouping. Transformations required to change a designated verdict included Register/Express inversion, signal/FSS type erasure, Store provenance deletion, target-coordinate omission, false-certificate insertion and generator substitution.

The lower-order nonfactorization corpus repeated every 12-bit observable vector with both higher-order labels. Logistic regression and a 200-tree random forest obtained held-out accuracies below 0.481. This check supports Eq. (6) of the Article for the finite paired corpus; it does not establish indistinguishability for every possible lower-order representation.

6 Synthetic world generator and intervention patches

6.1 Regime coordinates

The final design used 128 scrambled Sobol points plus 32 corner regimes. Fourteen coordinates controlled:

1. higher-order target prevalence;
2. bridge difficulty;
3. certificate noise;
4. Store noise;
5. Recall noise;
6. boundary noise;
7. Model residual;
8. Evaluate residual;
9. governance conflict;
10. hazard intensity;
11. realization fidelity;
12. option cost;
13. context demand;
14. target drift.

A target was higher-order when the first coordinate was at least 0.30. Every regime used 16 participants and eight common-random-number replicates per intervention.

6.2 Paired nonfactorizing target

For every lower-order projection q_n , the world generator pairs two worlds satisfying

$$q_n(W^0) = q_n(W^1), \quad \Phi_{n+1}(W^0) \neq \Phi_{n+1}(W^1). \quad (6)$$

A full solution requires grounded Register, active Recall, admissible promotion, span, certificate, Store, recursive reuse and Express. If the composition is unavailable, the generator emits a no-leakage guess internally; output failure produces no correct externally expressed verdict.

6.3 Ten conditions

All conditions are versioned patches to one generator.

1. local order-preserving baseline;
2. matched compute increase;
3. matched memory/context increase;
4. matched population/bridge-attempt increase;

5. matched communication increase;
6. certification improvement without lift;
7. operand lift without certified global span;
8. span without operand lift;
9. full CNS: boundary, recovery, lift, span, certificate, residual closure and participant governance;
10. residual-open CNS: the same nominal capability family with certificate, boundary, governance and target-shift residuals outside the full budget.

Resource counters are reported in Table 7. Exact matching is impossible because some contracts require validation or communication work by definition; the benchmark reports the unmatched dimensions rather than silently assigning them zero cost.

7 Integrated CNS results

On higher-order worlds, full CNS target accuracy was 0.585 and certified reach 0.435. The best matched target accuracy was 0.400; best non-CNS certified reach was 0.0033. Recursive reuse under full CNS was 0.347. Full numerical results are in Table 5. On local worlds, full CNS accuracy was 0.679 and matched compute accuracy was 0.595; the higher-order advantage therefore contracted as preregistered (Table 6).

The interaction is structural. Lift-only can construct a composite but cannot certify a remote target path. Span-only can form nominal cross-region paths but lacks the operand on which the higher-order predicate is defined. Certification-only can validate existing edges but cannot create either missing factor. Full CNS closes the conjunction; the residual-open condition shows that capability can be present while certified reach and output fidelity remain lower.

8 Option containment, governance and hazard projection

Proposition 3 (Participantwise finite option bound). *Let $O_b \subseteq O_a$. For participant i , suppose represented and realized fixed-target values differ by at most ϵ_i , selection error is at most δ_i , every baseline option remains admissible, and governance selects a new option only if its represented lower bound is no worse than the best baseline. Then*

$$u_i(a^*) \geq \max_{b \in O_b} u_i(b) - (2\epsilon_i + \delta_i). \quad (7)$$

Proof. Let $\hat{u}_i$ be represented value. Selection gives $\hat{u}_i(a^*) \geq \max_{b \in O_b} \hat{u}_i(b) - \delta_i$. Bounded representation error gives $u_i(a^*) \geq \hat{u}_i(a^*) - \epsilon_i$ and $\hat{u}_i(b) \geq u_i(b) - \epsilon_i$. Combining yields the claim. $\square$

The full CNS selected the new option in 0.099 of trajectories and improved mean outcome over the retained baseline by 0.015. Participant containment within -0.03 held in 0.987; the participant-action fraction strictly below baseline was 0.012. The residual-open control selected a new option in 0.366, had similar mean gain 0.014, but containment fell to 0.717 and harm rose to 0.156. The difference is not generated by the capability label: it follows the stored certificate, projection, realization and governance conditions.

The hazard view maps generated consequence intensity, unresolved residual, target failure and realization fidelity to a terminal-risk sample. Frequencies were 0.207 for full CNS, 0.266 for the residual-open control and 0.320 for baseline. These values are internal to the synthetic bridge. They do not estimate extinction probability.

9 Bounded registration and correlation-aware recovery

Definition 6 (Target adequacy). *For complete state X , projection $q : X \rightarrow Z$ and target Φ , q is exact for Φ iff*

$$q(x) = q(y) \implies \Phi(x) = \Phi(y). \quad (8)$$

Equivalently, $\Phi = f \circ q$ for some decoder f .

If all L relations are necessary and retained independently with common probability q , exact recovery is q^L . The experiment intentionally violates homogeneity by varying relation type, pipeline, budget, redundancy and a latent record-quality term. The grid contains:

- $L \in \{2, 4, 6, 8, 12, 16\}$;
- $K \in \{128, 256, 512, 1024\}$ equivalent fields/tokens;
- redundancy $r \in \{1, 2, 4\}$;
- generic summary, relation extraction and target-aware three-valued monitoring;
- three synthetic trace families and 30 tasks per cell.

Table 8 reports $K = 256, r = 1$. Generic exact recovery falls from 0.300 at $L = 2$ to 0.000 at $L = 16$. The target-aware monitor reaches 0.867 at $L = 2$ and 0.067 at $L = 8$ with false confidence 0.000. Its advantage is partly abstention: an incomplete critical graph returns UNDETERMINED.

Held-out log loss was 0.258 for the correlation-aware model and 1.024 for homogeneous independence. Thus the homogeneous q^L model is rejected as a quantitative description of these records, while capacity- and structure-dependent exact recovery remains strongly supported.

The external trace phase was not executed. No claim is made that these synthetic families are samples from current language models.

10 Model codewords and held-out identification

Let the candidate library be $\mathfrak{M} = \{M^*, M_1, \dots, M_{12}\}$. The load-bearing divergences are:

Table 1: Candidate reconstruction library.

ID	Reconstruction	Divergence
M^*	Canonical v8	Non-FSS carriers, Register/Express, Store/Recall/ L/G , promotion, three fitness coordinates, six functions, certificate, participant governance and one generator.
M_1	Signal-less	External structure is treated as already conceptual; Register/Express absent.
M_2	Signal regress	Carrier is treated as an FSS requiring another boundary.
M_3	Memory collapse	Store and Recall supplied by one undifferentiated or external index.
M_4	Verdict-only L/G	Global labels do not preserve evidence and successor obligations.
M_5	Connectivity-only G	Generic connectivity replaces obligation-preserving global assessment.
M_6	Primitive promotion	Promotion is not the registered composite Create/promotion contract.
M_7	No target coordinate	Current and projected fitness retained; adequacy supplied externally.
M_8	No projection coordinate	Current and target retained; action consequences not internally represented.
M_9	Aggregate governance	Participant fitness collapsed to one mean.
M_{10}	Uncertified reach	Nominal reach counted without certificate gating.
M_{11}	Split generator	Modules use separate fitted transitions.
M_{12}	Target-shift value	Target movement counted as achieved improvement.

For selected test set Q , model M_j predicts codeword

$$\mathbf{y}(M_j) = (q_1(M_j), \dots, q_K(M_j)). \quad (9)$$

The selector retained 27 tests after preserving at least one test from each causal layer. Under 1.5% bit flips, canonical identification was 0.940 (Wilson interval 0.916–0.958); the minimum candidate accuracy was 0.932. Table 9 reports every candidate. Eight held-out contract recombinations had selected-codeword distance at least 3 from M^* .

Theorem 2 (Finite codeword identification). *If every non-equivalent candidate codeword lies at distance greater than $2e$ from $\mathbf{y}(M^*)$ and the observed error has Hamming radius at most e , nearest-valid-codeword decoding uniquely returns M^* .*

Proof. The observed word lies within e of $\mathbf{y}(M^*)$. If another candidate were also within e , the triangle inequality would place the two candidate codewords within $2e$, contradicting the premise. $\square$

The theorem applies only to the frozen finite candidate class and error radius. A new target-different model sharing the selected signature defeats uniqueness and requires a larger test pool.

11 Artifact reconstruction and propagation status

The intended transmitted package is

$$\mathcal{A}_{\text{pkg}} = (A, S, \Omega^*, C, D, T, \mathbf{y}^*), \quad (10)$$

where A is the Article, S this Supplement, Ω^* the registry/projection table, C source code, D event data, T conformance and witness tests, and $\mathbf{y}^*$ the observed signature.

A reconstruction is operationally equivalent when an admissible renaming preserves layer types, function contracts, carrier/FSS domains, transition order, three-dimensional fitness geometry, theorem dependencies, primary predictions and held-out target behavior. Linguistic similarity is neither necessary nor sufficient.

The algorithmic stress test assigned lexical, schema-first and conformance-driven parsers to six package conditions. The complete package produced mean pass 0.856 (Wilson interval 0.768–0.914) compared with 0.000 for Article-only. Full results are in Table 10. The negative control was rejected in 0.711 of runs.

The preregistered independent decision rule required: no systematic fatal inversion; exact mandatory contract recovery; at least 95% deterministic prediction; at least 90% unseen-extension success; complete-package lower 95% bound above 0.80; improvement of at least 0.25 over Article-only; a positive result-signature increment; and independent receiver families. The present pooled lower bound was 0.768, and the receivers were internal parsers. Therefore

$$\text{PropagationVerdict} = \text{UNDETERMINED}. \quad (11)$$

This status localizes a transmission residual. It does not reverse the deterministic or integrated synthetic results, and those results do not upgrade propagation by implication.

12 Statistical plan and uncertainty

The master seed was 20260715. A runtime-only pilot determined that 128 Sobol plus 32 corner regimes and eight replicates could be executed with complete event logging in the available environment. Counts were frozen before the final labelled output was inspected. The resulting maximum Monte Carlo standard error for a probability near 0.5 is approximately 0.014 within the higher-order per-condition sample; intervals are descriptive numerical uncertainty, not estimates for human populations.

Primary stochastic summaries use mean ± 1.96 Monte Carlo standard errors clipped to observed support. Identification and receiver pass intervals use Wilson intervals[3]. The five planned result families were not converted into a familywise significance claim because the independent receiver and external trace families were not executed. Deterministic contract failures remain failures irrespective of a stochastic mean.

13 Counterexample and non-dominance regimes

The benchmark preregisters regimes in which a CNS should not dominate or should return UNDETERMINED:

- no higher-order target;
- failed lift, span, Store or Recall;
- ungrounded Register or failed Express;
- certificate residual outside budget;
- context too small to form the composition;
- comparator resources sufficient to enumerate the target;
- capability cost exceeding benefit;
- participant-incomplete governance;

- non-embeddable target order without a valid partial-order controller;
- ungoverned target revision;
- realization failure or hazard increase dominating recognition benefit;
- a prior innovation matching the full historical profile;
- artifact packages that fail to regenerate an equivalent model.

These regimes prevent the model label from predetermining positive valence or historical rank.

14 Claim-status ledger

Table 2: Status of principal claims after execution.

Claim	Status	Stronger requirement
Signal carriers are non-FSS typed objects and observable FSS interaction needs the declared input/output pair $\{\text{Store}, \text{Recall}, L, G\}$ is irreducible	Theorem within target class; conformance pass	A broader target class or empirical interface mapping.
Current, target and projected fitness are necessary and sufficient	Constructive result for the witness family	Independent target families without hidden resources.
Lift plus certified span generates non-factorizing reach	Exact paired-witness result	Controllers with broader partial-order targets and candidate fourth distinctions.
Residual-closed option expansion preserves participant bounds	Synthetic integrated result	Replication in independently generated worlds or empirical tasks.
Finite projections lose exact critical structure	Conditional synthetic result	Measured projection, governance and realization error bounds.
Joint signature identifies M^*	Synthetic and analytic result	External trace data from version-locked model families.
The package is propagation-closed	Finite candidate-class result	Independent held-out models and larger error budgets.
A CNS is historically maximal or civilization’s highest-importance innovation	Undetermined	Independent semantic receiver study meeting every conjunction threshold.
	Undetermined	Frozen full-profile historical audit with independent evidence.

15 Complete generated registry and result tables

The following tables are emitted directly from the registry and executed data files.

Table 3: Canonical function registry.

ID	Function	Layer	Contract
B01	Register	boundary	admit typed external signal structure into an FSS
B02	Express	boundary	encode selected FSS structure onto a typed external carrier
C01	Store	conceptual	preserve recoverable identity, relation, verdict, and provenance
C02	Recall	conceptual	reactivate a stored object under a declared target and budget
C03	L	conceptual	evaluate or transform within the active represented region
C04	G	conceptual	evaluate or transform across represented regions while preserving successor obligations
C05	CreatePromotion	conceptual	promote an admissible composite to a reusable higher-order operand

ID	Function	Layer	Contract
F01	Model	fitness	represent action-conditioned participant consequences
F02	Evaluate	fitness	compare represented consequences with participant targets and viability
F03	Stability	fitness	preserve a viable trajectory when correction is unnecessary
F04	Adaptation	fitness	change trajectory when represented deviation requires correction
F05	Decomposition	fitness	factor a failed target into typed residual obligations
F06	Bridging	fitness	construct an admissible correction spanning decomposed obligations
X01	Certificate	coupling	validate a target-relevant path and its boundary conditions
X02	Governance	coupling	authorize participant-complete action selection and target revision

Table 4: Deterministic target-pair witnesses. The complete model passed every row.

ID	Contract	Target distinction	Ablation	Typed reason
W01	Register	grounded external symbol must become a typed conceptual node	fail	input signal remains outside the conceptual state
W02	Express	stored verdict must be reconstructed by an external receiver	fail	no output carrier contains the verdict
W03	Store	identity must persist after active context eviction	fail	identity and provenance are lost
W04	Recall	stored bridge must reactivate under a later target	fail	stored object remains inert
W05	L	local relation classification from active evidence	fail	local evidence is not evaluated
W06	G	cross-region support must preserve all successor obligations	undetermined	local verdict cannot certify the global target
W07	current fitness	detect present viability violation with equal target and projection	fail	two states collapse without current coordinate
W08	target fitness	distinguish adequacy under equal current and projected outcomes	fail	two goals collapse without target coordinate
W09	projected fitness	select between actions with equal current and target	fail	action consequences collapse without projected coordinate
W10	Model	represent action-conditioned consequences	undetermined	consequences are unavailable
W11	Evaluate	compare projected state with target and viability	undetermined	no target-relative verdict
W12	Stability	preserve a viable action against unnecessary perturbation	fail	controller changes a viable trajectory
W13	Adaptation	change action after a target-relevant deviation	fail	controller remains on a failing trajectory
W14	Decomposition	localize a conjunctive failure to its typed obligations	undetermined	residual cannot be assigned
W15	Bridging	construct a correction joining separated obligations	fail	obligations remain disconnected

ID	Contract	Target distinction	Ablation	Typed reason
W16	CreatePromotion	promote an admissible composite as a reusable operand	undetermined	higher-order target remains outside operand domain
W17	operand lift	distinguish paired worlds with identical lower-order projections	undetermined	pair remains observationally equivalent
W18	certified span	connect lifted operand to a remote target with certificate	fail	lifted object has no reliable path
W19	Certificate	reject a decoy bridge with identical surface type	fail	nominal reach is mistaken for certified reach
W20	Governance	reject mean-positive action that violates one participant viability region	fail	participant harm is averaged away
W21	signal carrier type	reject recursive assignment of an internal FSS basis to a carrier	fail	type regress is admitted
W22	target revision gate	separate target change from achieved progress	fail	target motion is counted as positive displacement
W23	split generator	detect a locally fitted surrogate on a held-out transition	fail	mixed transition semantics pass unnoticed

Table 10: Algorithmic artifact-reconstruction stress test.

Receiver	Package	Contract recovery	Exact recovery	Extension	Pass
conformance_parser	article_only	0.300	0.000	0.367	0.000
conformance_parser	article_supplement	0.380	0.000	0.467	0.000
conformance_parser	complete_package	0.998	0.967	0.867	0.833
conformance_parser	negative_control	1.000	1.000	0.933	0.867
conformance_parser	plus_code_tests	0.948	0.333	0.900	0.333
conformance_parser	plus_registry	0.683	0.000	0.733	0.000
lexical_parser	article_only	0.500	0.000	0.400	0.000
lexical_parser	article_supplement	0.548	0.000	0.367	0.000
lexical_parser	complete_package	1.000	1.000	1.000	0.967
lexical_parser	negative_control	0.985	0.767	0.967	0.333
lexical_parser	plus_code_tests	0.712	0.000	0.800	0.000
lexical_parser	plus_registry	0.673	0.000	0.600	0.000
schema_parser	article_only	0.377	0.000	0.367	0.000
schema_parser	article_supplement	0.455	0.000	0.467	0.000
schema_parser	complete_package	0.995	0.900	0.933	0.767
schema_parser	negative_control	0.998	0.967	1.000	0.733
schema_parser	plus_code_tests	0.957	0.567	0.967	0.567
schema_parser	plus_registry	0.832	0.033	0.833	0.033

16 Reproducibility manifest

The public package contains:

- `run_experiment.py`: canonical generator, worlds, views and figure builder;
- `registry/functions.json`, `aliases.json`, `obligations.json`;
- model and event JSON schemas;

Table 5: Higher-order paired worlds.

Condition	Target acc.	Certified reach	Reuse	Containment	Harm
local_baseline	0.311	0.000	0.000	1.000	0.000
matched_compute	0.346	0.000	0.000	1.000	0.000
matched_memory	0.316	0.000	0.000	1.000	0.000
matched_population	0.333	0.000	0.000	1.000	0.000
matched_communication	0.400	0.000	0.000	1.000	0.000
certification_only	0.327	0.003	0.000	1.000	0.000
lift_only	0.310	0.000	0.000	1.000	0.000
span_only	0.317	0.000	0.000	1.000	0.000
full_cns	0.585	0.435	0.347	0.993	0.007
incoherent_cns	0.397	0.160	0.130	0.713	0.156

Table 6: Local-target worlds.

Condition	Target acc.	Certified reach	Reuse	Containment	Harm
local_baseline	0.468	0.000	0.000	1.000	0.000
matched_compute	0.595	0.000	0.000	1.000	0.000
matched_memory	0.522	0.000	0.000	1.000	0.000
matched_population	0.476	0.002	0.000	1.000	0.000
matched_communication	0.571	0.002	0.000	1.000	0.000
certification_only	0.478	0.004	0.000	1.000	0.000
lift_only	0.498	0.000	0.000	1.000	0.000
span_only	0.552	0.000	0.000	1.000	0.000
full_cns	0.679	0.472	0.397	0.976	0.022
incoherent_cns	0.554	0.172	0.125	0.724	0.155

Table 7: Mean recorded resources per trajectory.

Condition	Compute	Memory	Communication	Validation	Operations
certification_only	6.00	2.50	3.20	5.30	8
full_cns	6.70	2.50	3.62	4.03	8
incoherent_cns	6.70	2.50	3.62	4.03	8
lift_only	6.00	2.50	3.20	3.50	8
local_baseline	6.00	2.50	3.20	3.50	8
matched_communication	6.00	2.50	5.00	3.50	8
matched_compute	9.00	2.50	3.20	3.50	8
matched_memory	6.00	2.50	3.20	3.50	8
matched_population	6.00	2.50	3.44	3.50	8
span_only	6.00	2.50	4.40	3.50	8

Table 8: Bounded-registration results at budget 256 and redundancy one, averaged across three synthetic trace families.

L	Pipeline	Exact recovery	Relation retention	False confidence
2	generic_summary	0.300	0.561	0.333
2	relation_extraction	0.700	0.822	0.144
2	target_aware_monitor	0.867	0.928	0.000
4	generic_summary	0.156	0.544	0.544
4	relation_extraction	0.211	0.656	0.411
4	target_aware_monitor	0.622	0.875	0.000
6	generic_summary	0.044	0.387	0.522
6	relation_extraction	0.056	0.531	0.411
6	target_aware_monitor	0.233	0.750	0.000
8	generic_summary	0.000	0.324	0.533
8	relation_extraction	0.022	0.417	0.511
8	target_aware_monitor	0.067	0.611	0.000
12	generic_summary	0.000	0.109	0.478
12	relation_extraction	0.000	0.152	0.267
12	target_aware_monitor	0.000	0.249	0.000
16	generic_summary	0.000	0.023	0.500
16	relation_extraction	0.000	0.041	0.233
16	target_aware_monitor	0.000	0.113	0.000

Table 9: Noise-aware nearest-codeword identification accuracy.

True model	Accuracy
M1_signal_less	1.000
M4_verdict_only_LG	1.000
M3_store_recall_collapse	0.996
M2_signal_regress	0.988
M5_connectivity_only_G	0.972
M7_no_target	0.964
M6_primitive_promotion	0.964
M11_split_generator	0.960
M8_no_projection	0.952
M_star	0.940
M10_uncertified_reach	0.938
M12_target_shift_value	0.934
M9_aggregate_governed	0.932

- frozen confirmatory configuration and dependency versions;
- compressed integrated trajectories and complete event log;
- deterministic witnesses, fitness-triad and nonfactorization results;
- bounded-observation raw and summary records;
- model codewords, selection manifest, identification runs and held-out variants;
- artifact-reconstruction raw and summary records;
- vector and raster figures;
- conformance and replay tests;
- Article and Supplement source.

Minimal execution is:

```
python run_experiment.py
python tests/test_conformance.py
pdflatex nature_article_v8.tex
bibtex8 nature_article_v8
pdflatex nature_article_v8.tex
pdflatex nature_article_v8.tex
pdflatex nature_supplement_v8.tex
bibtex8 nature_supplement_v8
pdflatex nature_supplement_v8.tex
pdflatex nature_supplement_v8.tex
```

Required Python packages and versions are listed in `requirements.txt`. The execution manifest stores file hashes and environment metadata. Reproducible computational practice follows the principle that every reported result should be regenerated from code and preserved inputs rather than manually edited tables[4].

17 External completion protocol

A future propagation-closure run should freeze this package, select at least three independent receiver-model families with browsing disabled, use fresh contexts and fixed budgets, and assign 30 independent reconstructions per family and package. Receivers should emit standardized JSON, map local inventories to the registry, predict 20 secret held-out tasks, complete an implementation adapter, solve an unseen topology/vocabulary extension generated after package freeze, and reject a fluent negative-control inversion. No author clarification is permitted after assignment; public conformance errors are permitted.

The external trace phase should collect at least three version-locked model families on graph reach, higher-order relation classification, theorem-premise tracking, program execution, multi-participant governance and signal round trips. It should report exact critical-subgraph recovery, graph edit distance, valid/invalid/undetermined accuracy, abstention calibration, false confidence, backward Store/Recall recovery and boundary provenance. These data would test the bridge from the synthetic bounded-registration mechanism to an external measured boundary.

A historical audit should freeze candidate innovations, evidence cut-off, ambiguity rules and the full profile before scoring writing, mathematics, institutions, programmable/networked computing, machine learning and additional candidates. Missing evidence must remain undetermined. Any prior innovation matching the full profile defeats an exclusive historical claim rather than being excluded post hoc.

References

- [1] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. doi: 10.1002/j.1538-7305.1948.tb01338.x.
- [2] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, 2 edition, 2006.
- [3] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.
- [4] Geir Kjetil Sandve, Anton Nekrutenko, James Taylor, and Eivind Hovig. Ten simple rules for reproducible computational research. *PLoS Computational Biology*, 9(10):e1003285, 2013. doi: 10.1371/journal.pcbi.1003285.